



RESEARCH & INNOVATION UPDATES

Flexible Solar Cells Achieve 24.6% Efficiency



Flexible perovskite/CIGS tandem solar cells developed using the antisolvent-seeding approach (Credit: NIMTE)

Researchers from the Chinese Academy of Sciences (Beijing, China) have improved the durability of flexible tandem solar cells by strengthening the bond between the top perovskite and bottom CIGS layers. CIGS, known for its tunable bandgap, light absorption, and thermal stability, is ideal for the bottom layer. To address surface roughness that hinders perovskite formation, the team used a two-step SAM process and solvent engineering, along with a perovskite seed layer for better adhesion. The resulting 1.09cm² cell achieved 24.6% efficiency and maintained over 90% performance after 3000 bends, advancing flexible solar technology toward commercialisation.

Low-Cost Optical Detection Using LDR

Scientists from Iran University of Science and Technology (Narmak, Tehran, Iran) have developed a low-cost optical detection method for microfluidic Lab-on-a-Disc (eLOD) devices using light-dependent resistors (LDRs). The system combines LDRs with LEDs and custom waveguides to monitor real-time fluid flow, droplet behaviour, and sample volume. Tested with dyed fluids and high-speed rotation, it tracked red blood cell separation and deformability, which is key for diagnosing diseases like sickle cell anaemia. This compact, affordable setup supports real-time analysis and may revolutionise point-of-care diagnostics, especially in resource-limited areas.

Optical detection using LDR sensors (Credit: <https://www.azosensors.com>)

